

MINOR PROJECT – 2

AWS Identity and Access Management (IAM)

Implementation, Access Control and Permission Validation

Student	Dalia Mahata
Project	AWS IAM
Platform	Amazon Web Services
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This report presents the IAM users, groups, policies and permission validation carried out in the AWS console.

1. Introduction

AWS Identity and Access Management (IAM) is used to manage authentication and authorization for AWS resources. In this project, different IAM users were assigned policies according to their responsibilities. Selected permissions were then tested using the IAM Policy Simulator.

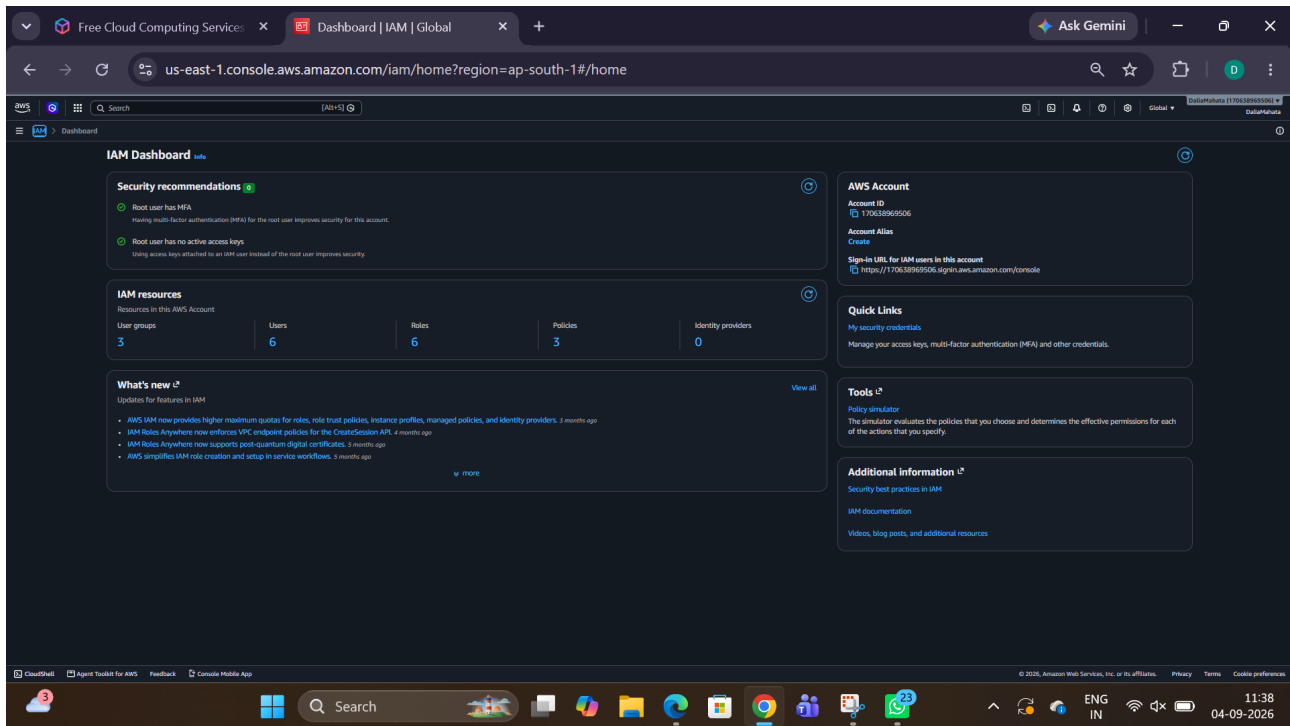
2. Objectives

- Create and configure IAM users and groups.
- Assign customer-managed and AWS-managed policies.
- Implement role-based access control.
- Test selected AWS actions using the Policy Simulator.
- Document successful, denied and resource-dependent validation results.

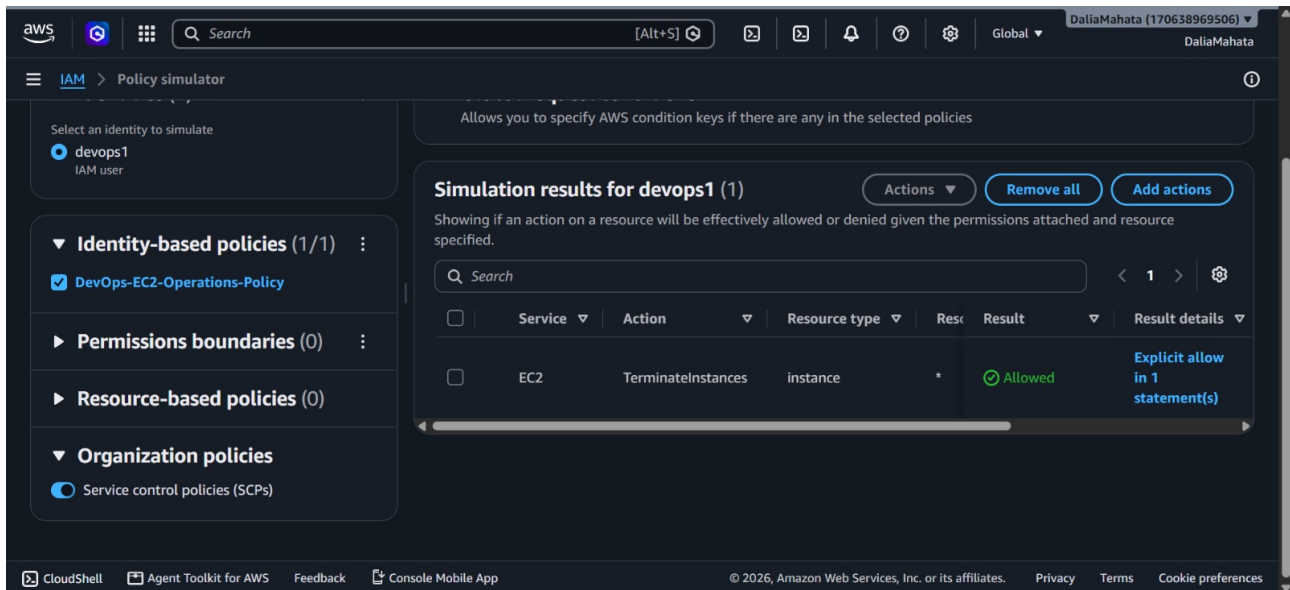
3. Project Configuration

User	Policy / Configuration
developer1	Developers-S3-Policy; attached directly and through Developers group
developer2	Developers-S3-Policy + IAMUserChangePassword
devops1	DevOps-EC2-Operations-Policy through DevOps group
auditor1	ReadOnlyAccess

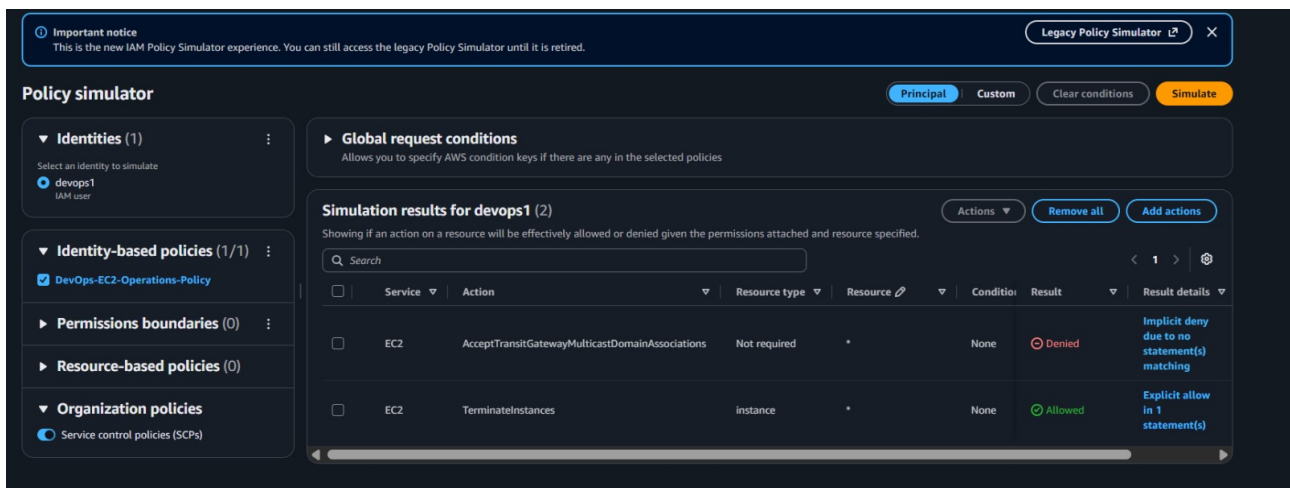
4. IAM Implementation Evidence



AWS IAM dashboard



developer2 — Permissions



5. Permission Simulator Results

The following tests were performed using the IAM Policy Simulator.

User	Action	Result
developer1	S3: PutObject	Denied — implicit deny in captured test
developer1	S3: DeleteBucket	Denied — implicit deny
devops1	EC2: TerminateInstances	Allowed — explicit allow
auditor1	EC2: RunInstances	Result blank; 5 resource types required

developer2

info

Delete

Summary

ARN

arn:aws:iam::170638969506:user/developer2

Created

September 01, 2026, 23:11 (UTC+05:30)

Console access

Enabled without MFA

Last console sign-in

Never

Access key 1

Create access key

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (2)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

All types

Search

< 1 >

☐

Policy name

☐

Developers-S3-Policy

☐

IAMUserChangePassword

▲

Type

▼

Customer managed

AWS managed

Attached via

Directly, Group Developers

Directly

▶ Permissions boundary (not set)

▼ Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

devops1 — EC2 TerminateInstances validation



6. Simulator Limitations

Some IAM actions require resource-specific values before the simulator can return a final decision. In the captured tests, S3 PutObject and EC2 RunInstances did not produce the intended final validation state. These outputs are recorded as simulator/resource-validation limitations and are not presented as proof that the underlying IAM configuration is incorrect.

7. Conclusion

The project demonstrates practical AWS IAM access control using users, groups and policies. Developers were given S3-related permissions, the auditor was configured with read-only access, and the DevOps user received EC2 operational permissions. The simulator successfully confirmed the explicit allow for devops1 to perform EC2 TerminateInstances and also showed denied actions for developer1. The collected screenshots provide implementation and validation evidence.

8. References

- Amazon Web Services — IAM Console
- AWS Identity and Access Management documentation
- AWS IAM Policy Simulator

— End of Report —